

MELBOURNE FERNY CREEK, AustraliaWMO: **948720**Lat: **37.8749S**Lon: **145.3500E**Elev: **513**StdP: **95.31**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.3	3.0	-3.0	3.1	14.5	-0.8	3.8	10.8	14.3	7.6	13.3	7.9	3.6	290	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	30.6	17.3	28.3	17.0	26.4	16.3	19.7	25.7	18.6	24.9	17.7	23.8	3.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	13.6	21.4	16.7	12.6	20.5	15.7	11.8	19.6	58.4	25.2	54.9	25.0	51.9	24.0	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	10.9	9.2	0.7	35.0	1.0	2.5	0.0	36.8	-0.6	38.2	-1.1	39.6	-1.9	41.4
			0.1	21.6	1.1	1.1	-0.6	22.4	-1.3	23.0	-1.9	23.6	-2.7	24.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	17.4	17.5	15.7	12.9	9.9	7.7	6.9	7.6	9.4	11.3	13.6	15.2
	DBStd	5.14	4.64	4.49	4.32	3.54	2.71	2.03	1.84	2.41	3.01	3.88	4.32	4.58
	HDD10.0	389	1	1	2	11	35	73	97	83	46	28	9	4
	HDD18.3	2446	76	65	108	168	261	318	353	333	267	221	154	122
	CDD10.0	1144	230	211	178	97	33	5	2	8	29	69	118	165
	CDD18.3	159	47	42	25	3	0	0	0	0	4	13	26	26
	CDH23.3	1320	456	339	164	8	0	0	0	0	1	19	92	242
	CDH26.7	404	174	99	41	0	0	0	0	0	0	2	13	74
(14) Wind	WSAvg	3.5	3.6	3.4	3.2	3.1	3.2	3.7	3.8	3.8	3.7	3.6	3.5	3.4
(15) Precipitation	PrecAvg	893	60	60	63	79	70	84	73	78	90	87	76	72
	PrecMax	1191	114	178	108	201	146	167	139	120	172	167	166	169
	PrecMin	559	13	4	34	26	18	24	30	26	33	27	28	7
	PrecStd	160	27	46	20	40	33	32	27	27	38	37	36	39
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	32.7	30.9	24.3	18.5	14.3	13.2	16.8	20.6	25.7	28.3	32.8
		MCWB	18.9	17.5	17.6	14.9	11.8	10.5	9.5	11.0	12.8	14.5	16.8	17.5
	2%	DB	30.8	29.5	27.3	21.8	16.7	12.8	11.6	14.4	18.2	22.5	26.0	28.7
		MCWB	17.6	17.1	16.4	14.3	11.6	10.0	8.4	9.8	12.0	13.4	15.6	16.6
	5%	DB	28.2	27.3	24.9	20.0	15.5	11.7	10.5	12.7	16.3	20.3	23.6	26.0
		MCWB	17.2	16.9	16.0	13.2	11.2	9.2	8.0	8.8	10.9	12.7	15.0	15.9
	10%	DB	25.5	25.2	22.5	18.5	14.2	10.7	9.6	11.3	14.6	18.1	21.4	23.5
		MCWB	16.7	16.3	15.2	12.7	10.6	8.6	7.5	8.0	10.1	11.7	14.2	15.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	20.8	19.3	16.6	13.9	11.7	10.6	11.9	14.2	16.1	19.2	19.8
		MCDB	27.7	26.2	25.6	21.9	16.5	12.9	11.6	14.7	17.8	21.2	24.7	25.7
	2%	WB	19.9	19.4	18.0	15.2	12.6	10.6	9.3	10.5	12.8	14.7	17.2	18.2
		MCDB	26.3	25.1	24.2	19.4	15.1	12.1	10.5	13.4	16.9	20.0	22.9	24.8
	5%	WB	18.7	18.3	17.0	14.3	11.7	9.7	8.5	9.3	11.6	13.6	16.1	17.1
		MCDB	25.3	24.1	22.3	18.2	14.4	11.0	9.7	12.0	15.4	18.6	21.1	23.7
	10%	WB	17.6	17.3	16.1	13.5	10.9	8.9	7.8	8.5	10.5	12.4	14.9	16.0
		MCDB	23.9	22.8	21.3	17.3	13.5	10.3	9.1	10.7	13.8	17.1	19.9	22.0
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	9.4	8.1	6.1	4.1	3.4	3.5	4.6	6.2	7.7	8.7	9.4
		MCDBR	14.0	12.5	11.5	8.4	5.4	4.3	4.4	6.4	8.4	10.8	12.0	13.2
	5% WB	MCWBR	5.7	5.1	5.0	4.1	3.5	3.2	3.0	3.7	4.6	5.5	5.9	5.9
		MCWBR	12.0	11.1	9.9	7.6	4.9	3.7	3.8	5.8	7.8	9.8	10.4	11.4
(40) Clear-Sky Solar Irradiance	taub		0.328	0.333	0.313	0.308	0.293	0.285	0.286	0.294	0.317	0.317	0.324	0.320
		taud	2.550	2.541	2.599	2.582	2.614	2.624	2.611	2.584	2.506	2.533	2.537	2.569
	Ebn at Noon		1005	977	958	900	858	837	857	904	936	983	1002	1018
		Edh at Noon	107	104	90	80	67	62	66	78	97	103	108	106
(44) All-Sky Solar Radiation	RadAvg		7.13	6.22	4.96	3.44	2.25	1.75	1.95	2.76	3.95	5.32	6.30	7.03
	RadStd		0.44	0.34	0.28	0.20	0.14	0.15	0.10	0.19	0.29	0.44	0.36	0.40

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.68	+0.44	N/A	N/A	N/A	N/A	N/A	-182	+190	+40
(47)	Regional (39 neighbors)	+0.50	N/A	N/A	+0.84	N/A	N/A	-17	-127	+173	+60

Nomenclature: See separate page